JC2001 Introduction to Software Engineering

Practical Session 4 - Exercises

Risk Analysis

Risk management in software engineering is a sequence of processes for identifying factors that may affect a project's success. These processes include risk identification, risk analysis, risk planning, and risk monitoring. Proper risk analysis helps to control possible future events that may harm the overall project. It is a proactive process rather than a reactive process. In this exercise, we will work on risk identification and analysis.

Exercise A: Risk Identification

Risk identification is a process of determining risks that could potentially prevent the project from achieving its objectives. It includes documenting and communicating the concern. Risk identification is the crucial first step of the risk management process. As a starting point for risk identification, a checklist of different types of risk may be used. Six types of risk may be included in a risk checklist:

- **Estimation risk:** This arises from the management estimation of the resources required to build the software system.
- **Organisational risk**: This arises from the organisational environment where the software is being developed.
- **People risk**: This is associated with the people in the development team.
- **Requirements risk:** This comes from changes to the customer's requirements and the process of managing the requirements change.
- **Technology risk**: This comes from the hardware or software technologies that are used to develop the system.
- **Tools risk:** This comes from the software tools and other support software used to develop the system.

Question 1: Explain the importance of risk management in an organisation.

Exercise B: Risk Analysis

In this stage of the risk management process, the likelihood and consequences of these risks are assessed. There are no hard or fast rules for judging the probability and seriousness of these risks. Instead, one uses experience from past projects and the problems that arose to assess the risk involved.

Question: You are leading a team of 2 software developers in implementing a new software product. The deadline for releasing an alpha-version is in two weeks. However, one developer decided to call in sick. 75% of the functions are completed. Is this a low, medium or high risk scenario? Discuss.

Exercise C: Case Study

Insulin pumps are portable devices attached to the body that continuously deliver rapid- or short-acting insulin via a catheter placed under the skin. An insulin pump is a small device that delivers insulin into the layer of fat that sits just below the skin. Because the insulin pump stays connected to the body, it allows the wearer to modify the amount of insulin they take with the press of a few buttons at any time of the day, or to program in a higher or lower rate of insulin delivery to occur at

a chosen time, which can be when sleeping. The UK insulin pump audit in 2013 showed that around 6% of adults with type 1 diabetes use an insulin pump. Although insulin pumps provide benefits, they also carry risks.

Question: Identify the risks associated with using an insulin pump and assign a severity level of low, medium or high to each risk.

References

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